Design of a Programmable Gain Amplifier Using a Fully Differential OTA with 5T-OTA CMFB and DC Offset Cancellation

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1 Design Specifications

1.1 OTA Specifications

Parameter	Target
Technology	SCL 180 nm CMOS
Supply voltage	1.8 V
Load capacitance	1 pF
DC open-loop gain	≥ 70 dB
-3 dB bandwidth	≥ 75 kHz
Phase margin	60°–80°
CMRR	≥ 80 dB
PSRR	≥ 65 dB
Slew rate	$\geq 20\,\mathrm{V}/\mu\mathrm{s}$
Settling time (2%)	≤ 200 ns
Power (OTA+CMFB)	≤ 1 mW
Input-referred noise	$\leq 35\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ @ 1 kHz
DC offset (after DCOC)	≤ 10 mV

1.2 PGA Specifications

Parameter	Target
Closed-loop gain range	10 dB – 50 dB
-3 dB BW at 50 dB gain	≥ 750 kHz
Gain error	$\leq 5\%$
Total power (incl. DCOC)	< 1.5 mW

1.3 CMFB Targets

- CMFB loop gain ≥ 40 dB
- CMFB UGF $\approx$ 5–10 MHz
- CMFB phase margin $\geq 60°$

2 Architecture Selection

2.1 Core OTA Topology

A two-stage fully differential OTA was selected primarily for headroom and gain reasons. With $V_{DD} = 1.8\,\mathrm{V}$ and only $\approx 200\,\mathrm{mV}$ overdrive per device, a folded cascode would stack 5–6 V_{ov} leaving insufficient headroom for the required 1.4 V output swing, whereas the two-stage topology requires only 4 stacked V_{ov}. Miller R_c–C_c compensation provides a clean, well-understood stability mechanism with a single zero-nulling resistor.

2.2 CMFB Topology

A 5T-OTA CMFB was chosen. It senses:

$$V_{CM,\mathrm{sense}} = \frac{V_{out+} + V_{out-}}{2} \tag{1}$$

via two equal resistors ($R_{\mathrm{sense}} = 500\,\mathrm{k\Omega}$) and compares to $V_{ref} = 0.9\,\mathrm{V}$. The error drives V_{cmfb} which adjusts current to restore CM balance.

2.3 Programmable feedback network

A switchable resistor ladder was chosen because gain accuracy depends only on resistor ratios, which track well across process corners. NMOS switches short ladder segments to select gain, keeping the signal path purely resistive.

2.4 DCOC

A low-frequency RC integrator loop was chosen for DC offset cancellation because it naturally implements a high-pass response, suppressing DC and near-DC offset while passing signals above $f_{HP} = 2\,\mathrm{kHz}$.

3 Extracted Process Parameters

Table 1: SCL 180 nm Process Parameters (Extracted)

Parameter	Symbol	Value	Notes
Min. channel length	$L_{\min}$	180 nm	
Min. channel width	$W_{\min}$	220 nm	
NMOS threshold	$V_{th,N}$	477.2 mV	
PMOS threshold		513.7 mV	At $V_{SB} = 0$
NMOS transconductance	$\mu_n C_{ox}$	234.5 μA/V^2	
PMOS transconductance	$\mu_p C_{ox}$	63.4 μA/V^2	
NMOS output resistance	$r_{o,N}$	149.8 kΩ	At $I_D = 125\,\mu$A, $L = 180$ nm
PMOS output resistance	$r_{o,P}$	108.1 kΩ	At $I_D = 125\,\mu$A, $L = 180$ nm
Gate oxide capacitance	C_{ox}	≈ 8.6 fF/μm^2	Estimated

4 Hand Calculations

4.1 (4a) Required Transconductance

The project specifies UGB using the relation for a Miller-compensated two-stage OTA:

$$\text{UGB} = \frac{g_{m1}}{2\pi C_c} \tag{2}$$

The PGA must achieve closed-loop -3 dB bandwidth ≥ 750 kHz at maximum gain of 50 dB ($G = 316$ V/V). The required OTA UGB is therefore:

$$\text{UGB} \geq G_{\max} \times BW_{CL} = 316 \times 750\,\text{kHz} = 237\,\text{MHz} \tag{3}$$

A practical target of **UGB = 53 MHz** is chosen, supporting closed-loop bandwidths up to ≈ 167 kHz at 50 dB gain (well above the 75 kHz open-loop spec).

Assumptions: $C_c = 3$ pF (derived from stability), $V_{ov1} = 150$ mV (moderate inversion for low noise).

Required g_{m1}:

$$g_{m1} = 2\pi \times \text{UGB} \times C_c = 2\pi \times 53 \times 10^6 \times 3 \times 10^{-12} = \boxed{1.00\ \text{mA/V}}$$

Simulation extracted: $g_{m1} = 912\,\mu A/V$ (within 9% of target).

4.2 (4b) Bias Current for the Differential Pair

From the square-law $g_m = 2I_D/V_{ov}$:

$$I_{D1} = \frac{g_{m1} \cdot V_{ov1}}{2} = \frac{1 \times 10^{-3} \times 0.150}{2} = 75\ \mu\text{A} \tag{4}$$

$$I_{tail} = 2 \times I_{D1} = 150\ \mu\text{A} \tag{5}$$

Slew rate verification (SR limited by C_c in Miller-compensated topology):

$$SR = \frac{I_{tail}}{C_c} = \frac{150 \times 10^{-6}}{3 \times 10^{-12}} = 50\ \text{V}/\mu\text{s} \geq 20\ \text{V}/\mu\text{s} \tag{6}$$

W/L sizing of M1/M2 (square law):

$$\left.\frac{W}{L}\right|_{M1} = \frac{g_{m1}^2}{2\,\mu_n C_{ox}\,I_{D1}} = \frac{(10^{-3})^2}{2 \times 234.5 \times 10^{-6} \times 75 \times 10^{-6}} = \frac{10^{-6}}{35.18 \times 10^{-9}} = 28.4 \tag{7}$$

With $L = 500\,\text{nm}$ (chosen for gain, see Sec. 5.3):

$$W_{M1} = 28.4 \times 500\,\text{nm} = 14.2\ \mu\text{m} \approx \mathbf{14}\ \mu\text{m} \tag{8}$$

Verification:

$$g_m = \sqrt{2 \times 234.5 \times 10^{-6} \times 28 \times 75 \times 10^{-6}} = 0.994\,\text{mA/V} \approx 1\,\text{mA/V}$$

$I_{D1} = I_{D2} = 75\,\mu\text{A}, \quad I_{tail} = 150\,\mu\text{A}, \quad W/L = 28, \quad L = 500\,\text{nm}, \quad W = 14\,\mu\text{m}$

4.3 (4c) Open-Loop Gain Estimate

Output Resistance Scaling

Stage Output Resistances

$$R_{out1} = r_{o1} \| r_{o3} = \frac{694\,\text{k} \times 360\,\text{k}}{694\,\text{k} + 360\,\text{k}} = 237\,\text{k}\Omega \tag{9}$$

$$R_{out2} = r_{o5} \| r_{o7} = \frac{469\,\text{k} \times 650\,\text{k}}{469\,\text{k} + 650\,\text{k}} = 272\,\text{k}\Omega \tag{10}$$

Stage Gains and Total DC Gain

$$A_1 = g_{m1} \cdot R_{out1} = 1\,\text{mA/V} \times 237\,\text{k}\Omega = 237\ \text{V/V} = 47.5\ \text{dB} \tag{11}$$

$$A_2 = g_{m5} \cdot R_{out2} = 0.8\,\text{mA/V} \times 272\,\text{k}\Omega = 218\ \text{V/V} = 46.8\ \text{dB} \tag{12}$$

$$A_0 = A_1 \times A_2 = 237 \times 218 = 51{,}666\ \text{V/V} \tag{13}$$

$$\boxed{A_0 = 20\log_{10}(51{,}666) = 94.3\ \text{dB} \geq 70\ \text{dB}} \tag{14}$$

Poles, Zero, and Phase Margin

Dominant pole (Miller effect on C_c):

$$f_{p1} = \frac{1}{2\pi\, g_{m5}\, R_{out2}\, R_{out1}\, C_c} = \frac{1}{2\pi \times 0.8\text{m} \times 272\text{k} \times 237\text{k} \times 3\text{p}} = 1.02\ \text{kHz} \tag{15}$$

Consistency check:

$$A_0 \times f_{p1} = 51{,}666 \times 1020 = 52.7\,\text{MHz} \approx \text{UGB}$$

Second pole:

$$f_{p2} = \frac{g_{m5}}{2\pi C_L} = \frac{0.8 \times 10^{-3}}{2\pi \times 10^{-12}} = 127.3\ \text{MHz} \tag{16}$$

RHP zero without R_c:

$$f_z = \frac{g_{m5}}{2\pi C_c} = \frac{0.8\,\text{mA/V}}{2\pi \times 3\,\text{pF}} = 42.4\ \text{MHz} \tag{17}$$

This zero is very close to UGB (53 MHz) and would degrade phase margin catastrophically. $R_c = 1/g_{m5}$ pushes it to infinity.

Phase margin:

$$\text{PM} = 90^\circ - \arctan\left(\frac{f_u}{f_{p2}}\right) = 90^\circ - \arctan\left(\frac{53.1}{127.3}\right) = 90^\circ - 22.6^\circ = \boxed{67.4^\circ} \tag{18}$$

M3/M4 Active Load Sizing

Choose $V_{ov3} = 200\,\text{mV}$, $L = 360\,\text{nm}$ (shorter L keeps mirror pole high):

$$g_{m3} = \frac{2I_{D3}}{V_{ov3}} = \frac{2 \times 75\,\mu}{0.200} = 750\ \mu\text{A/V} \tag{19}$$

$$\left.\frac{W}{L}\right|_{M3} = \frac{(750\,\mu)^2}{2 \times 63.4\,\mu \times 75\,\mu} = 59.1 \quad \Rightarrow \quad W_{M3} = 59.1 \times 360\,\text{nm} = 21.3\ \mu\text{m} \tag{20}$$

Mirror pole check ($C_{ox} \approx 8.6\,\text{fF}/\mu\text{m}^2$):

$$C_{gs3} \approx \tfrac{2}{3} W_3 L_3 C_{ox} = \tfrac{2}{3} \times 21\,\mu \times 360\,\text{n} \times 8.6\,\text{f} = 43.3\ \text{fF} \tag{21}$$

$$f_{\text{mirror}} = \frac{g_{m3}}{2\pi C_{gs3}} = \frac{750\,\mu}{2\pi \times 43.3\,\text{f}} = 2.76\ \text{GHz} \gg 53\ \text{MHz} \tag{22}$$

M5/M6 Second Stage Sizing

Choose $V_{ov5} = 200\,\text{mV}$, $L = 500\,\text{nm}$, $I_{D5} = 80\,\mu\text{A}$:

$$g_{m5} = \frac{2 \times 80\,\mu}{0.200} = 800\ \mu\text{A/V} \tag{23}$$

$$\left.\frac{W}{L}\right|_{M5} = \frac{(800\,\mu)^2}{2 \times 63.4\,\mu \times 80\,\mu} = 63.1 \quad \Rightarrow \quad W_{M5} = 63.1 \times 500\,\text{nm} = 31.6\ \mu\text{m} \tag{24}$$

M7/M8 Output Current Source Sizing

Choose $V_{ov7} = 150\,\text{mV}$, $L = 500\,\text{nm}$, $I_{D7} = 80\,\mu\text{A}$:

$$\left.\frac{W}{L}\right|_{M7} = \frac{2 \times 80\,\mu}{234.5\,\mu \times (0.150)^2} = 30.3 \quad \Rightarrow \quad W_{M7} = 30.3 \times 500\,\text{nm} = 15.2\ \mu\text{m} \tag{25}$$

M0 Tail Current Source Sizing

Choose $V_{ov0} = 200\,\text{mV}$, $L = 500\,\text{nm}$:

$$\left.\frac{W}{L}\right|_{M0} = \frac{2 \times 150\,\mu}{234.5\,\mu \times (0.200)^2} = 32.0 \quad \Rightarrow \quad W_{M0} = 32.0 \times 500\,\text{nm} = 16\ \mu\text{m} \tag{26}$$

Compensation Network

$$C_c = \frac{g_{m1}}{2\pi \times \text{UGB}} = \frac{10^{-3}}{2\pi \times 53 \times 10^6} = 3.0 \text{ pF} \tag{27}$$

$$R_c = \frac{1}{g_{m5}} = \frac{1}{901.86 \times 10^{-6}} = \mathbf{1.11\ k\Omega} \tag{28}$$

Headroom Verification

$$V_{CM,in,\min} = V_{ov0} + V_{th,N} + V_{ov1} = 200\,\text{m} + 477\,\text{m} + 150\,\text{m} = 827 \text{ mV} \tag{29}$$

$$\Rightarrow V_{CM,in} = 900 \text{ mV} > 827 \text{ mV} \tag{30}$$

$$V_{out,\max} = V_{DD} - |V_{ov5}| = 1.8 - 0.200 = 1.60 \text{ V} \tag{31}$$

$$V_{out,\min} = V_{ov7} = 150 \text{ mV} \tag{32}$$

$$\text{Swing} = 1.45 \text{ V} \tag{33}$$

4.4 (4d) CMFB Loop Bias Calculations

The CMFB 5T-OTA (M9–M13) must achieve loop gain $\geq 40\,\text{dB}$ and UGF = 5–10 MHz.

CMFB bias current (set to $\frac{1}{4}$ of main tail to keep CMFB slower than differential loop):

$$I_{tail,CMFB} = \frac{150\,\mu}{4} = 37.5\ \mu\text{A}, \quad I_{D10} = \frac{37.5\,\mu}{2} = 18.75\ \mu\text{A} \tag{34}$$

CMFB r_o at $I_D = 18.75\,\mu$A:

$$r_{o,N}^{CMFB} = 149.8\,\text{k} \times \frac{125}{18.75} \times \frac{500}{180} = 2.775 \text{ M}\Omega \tag{35}$$

$$r_{o,P}^{CMFB} = 108.1\,\text{k} \times \frac{125}{18.75} \times \frac{360}{180} = 1.441 \text{ M}\Omega \tag{36}$$

$$R_{out,CMFB} = 2.775\,\text{M} \| 1.441\,\text{M} = 950 \text{ k}\Omega \tag{37}$$

Required g_{m10} for $\geq 40\,$dB loop gain:

$$g_{m10,\min} = \frac{100}{R_{out,CMFB}} = \frac{100}{950\,\text{k}} = 105\ \mu\text{A/V} \tag{38}$$

Actual g_{m10} with $V_{ov10} = 150\,$mV:

$$g_{m10} = \frac{2 \times 18.75\,\mu}{0.150} = 250\ \mu\text{A/V} \gg 105\ \mu\text{A/V} \tag{39}$$

CMFB loop gain:

$$L_{CMFB} = g_{m10} \times R_{out,CMFB} = 250\,\mu \times 950\,\text{k} = 237.5\ \text{V/V} = \boxed{47.5\ \text{dB}} \tag{40}$$

CMFB UGF and load capacitor sizing:

Without extra cap, the intrinsic load is $C_{load} = 2C_{gs3} = 86.6\,\text{fF}$:

$$f_{u,CMFB,\text{no cap}} = \frac{250\,\mu}{2\pi \times 86.6\,\text{f}} = 459\ \text{MHz} \gg 10\ \text{MHz} \tag{41}$$

An explicit load cap is required at the V_{cmfb} node:

$$C_{L,CMFB} = \frac{g_{m10}}{2\pi \times 8\ \text{MHz}} = \frac{250\,\mu}{2\pi \times 8 \times 10^6} = 4.97\ \text{pF} \approx \mathbf{5}\ \text{pF} \tag{42}$$

$$f_{u,CMFB} = \frac{250\,\mu}{2\pi \times 5\,\text{pF}} = 7.96\ \text{MHz} \tag{43}$$

CMFB transistor sizing:

M10/M11 (NMOS, $V_{ov} = 150\,\text{mV}$, $L = 500\,\text{nm}$):

$$\left.\frac{W}{L}\right|_{M10} = \frac{(250\,\mu)^2}{2 \times 234.5\,\mu \times 18.75\,\mu} = 7.1 \quad \Rightarrow \quad W = 3.6\ \mu\text{m} \tag{44}$$

M12/M13 (PMOS, $V_{ov} = 200\,\text{mV}$, $L = 360\,\text{nm}$):

$$g_{m12} = \frac{2 \times 18.75\,\mu}{0.200} = 187.5\ \mu\text{A/V}, \quad \left.\frac{W}{L}\right|_{M12} = \frac{(187.5\,\mu)^2}{2 \times 63.4\,\mu \times 18.75\,\mu} = 14.8 \quad \Rightarrow \quad W = 5.3\ \mu\text{m} \tag{45}$$

M9 (NMOS tail, $V_{ov} = 200\,\text{mV}$, $L = 500\,\text{nm}$):

$$\left.\frac{W}{L}\right|_{M9} = \frac{2 \times 37.5\,\mu}{234.5\,\mu \times (0.200)^2} = 8.0 \quad \Rightarrow \quad W = 4.0\ \mu\text{m} \tag{46}$$

CM sensing resistors:

$$R_{sense} \gg R_{out2} = 272\ \text{k}\Omega \quad \text{and} \quad f_{pole} = \frac{1}{2\pi R_{sense} C_{gs10}} \gg f_{u,CMFB} \tag{47}$$

At $R_{sense} = 500\,\text{k}\Omega$:

$$f_{pole} = \frac{1}{2\pi \times 500\,\text{k} \times 10.3\,\text{f}} = 30.9\,\text{MHz} \gg 7.96\,\text{MHz}$$

The large ratio $\mu_n/\mu_p \approx 3.7$ means PMOS devices require roughly 3.7× larger W/L to achieve the same g_m at the same current. This significantly impacts the sizing of M3/M4 and M5/M6. Furthermore, the r_o values at minimum length are moderate — to achieve the $\geq 70\,\text{dB}$ gain target, channel lengths of 360–500 nm were necessary for all signal-path devices.

4.5 (4e) DC Offset Cancellation Filter Sizing and DCOC OTA Design

The DCOC loop implements a high-pass filter via an RC integrator in the feedback path. The high-pass cutoff f_{HP} controls the tradeoff between offset suppression and distortion of low-frequency signals.

Choice: $f_{HP} = 2\,\text{kHz}$

- Low enough to suppress DC drift and very low-frequency offset.
- High enough to preserve signal content above 2 kHz (sufficient for most baseband applications).

Time constant:

$$\tau = R_{OC}C_{OC} = \frac{1}{2\pi f_{HP}} = \frac{1}{2\pi \times 2 \times 10^3} = 79.6\ \mu\text{s} \tag{48}$$

Component values:

$$C_{OC} = 10\ \text{pF} \quad \text{(on-chip feasible)}, \quad R_{OC} = \frac{\tau}{C_{OC}} = \frac{79.6\,\mu}{10\,\text{p}} = 7.96\ \text{M}\Omega \approx \mathbf{8\ M\Omega} \tag{49}$$

4.6 (4f) MOS Switches, Body Effect, and Complete Sizing Table

PGA Switch Sizing

The PGA feedback network uses NMOS switches. The on-resistance must satisfy:

$$R_{on} \leq 5\% \times R_{seg,\text{min}} = 0.05 \times 31.6\ \text{k}\Omega = 1.58\ \text{k}\Omega \tag{50}$$

With $V_{GS} = V_{DD} = 1.8\,\text{V}$ and $V_{th,N} = 477\,\text{mV}$:

$$V_{ov,sw} = 1.8 - 0.477 = 1.323\ \text{V} \tag{51}$$

$$\left.\frac{W}{L}\right|_{sw} = \frac{1}{R_{on} \times \mu_n C_{ox} \times V_{ov,sw}} = \frac{1}{1580 \times 234.5 \times 10^{-6} \times 1.323} = 2.04 \tag{52}$$

With $L = L_{\min} = 180\,\text{nm}$: $W = 2.04 \times 180\,\text{nm} = 367\,\text{nm}$. Use $W = 2\,\mu\text{m}$ with body-effect margin.

Note on body effect: At mid-rail signal ($V_{SB} \approx 0.9\,\text{V}$), the effective threshold rises by $\Delta V_{th} \approx 225\,\text{mV}$, increasing R_{on}. In simulation, $R_{on} = 22.1\,\text{k}\Omega$ was measured at minimum width — requiring a scale-up:

$$W_{sw,corrected} = W_{old} \times \frac{R_{on,old}}{R_{on,target}} = 220\,\text{n} \times \frac{22.1\,\text{k}}{1.58\,\text{k}} = 3.1\ \mu\text{m} \tag{53}$$

Use $W_{sw} = \mathbf{4}\,\mu\text{m}$, $L = 180\,\text{nm}$ to ensure gain error $\leq 5\%$ across all gain settings.

Body Effect on M1/M2

M1/M2 sources sit at $V_{tail} \approx V_{ov0} = 200\,\text{mV}$ above GND, so $V_{SB} \approx 200\,\text{mV}$:

$$\Delta V_{th} \approx \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right) \approx 0.5 \times (\sqrt{0.8} - \sqrt{0.6}) \approx 50\ \text{mV} \tag{54}$$

This is accounted for by the conservative choice of $V_{ov1} = 150\,\text{mV}$ (larger than strictly needed from g_m alone) and verified in simulation.

Why $L > L_{\min}$ for All Analog Devices

Using $L = 180\,\text{nm}$ for gain-critical devices produces r_o values at the table reference, giving $A_0 \approx 65\,\text{dB}$ — which fails the 70 dB target. Increasing L to 500 nm raises r_o by $\times 2.78$, pushing A_0 to $\approx 94\,\text{dB}$ with comfortable margin. In simulation, this was confirmed: with $L = 180\,\text{nm}$ for all devices, $A_0 = 72\,\text{dB}$ (barely passing); after changing to $L = 500/360\,\text{nm}$, $A_0 = 94\,\text{dB}$.

5 Implementation Details

5.1 Complete Transistor Sizing Table

Table 2: Final Transistor Sizes — Core OTA and CMFB

Device	**Type**	**W (μm)**	**L (nm)**	**W/L**	I_D **(μA)**	V_{ov} **(mV)**	g_m **(μA/V)**	**Role**
M1, M2	NMOS	14	500	28	71	150	912	Input diff p
M3, M4	PMOS	21	360	59	71	200	750	Active load
M5, M6	PMOS	32	500	63	71	200	902	Second stag
M7, M8	NMOS	15	500	30	71	150	948	Output cur
M0	NMOS	16	500	32	142	200	—	Tail current
M9	NMOS	4	500	8	37.5	200	—	CMFB tail
M10, M11	NMOS	3.6	500	7	18.75	150	250	CMFB sens
M12, M13	PMOS	5.3	360	15	18.75	200	188	CMFB acti

Table 3: Passive and Switch Values

Component	**Value**	**Basis**
C_c (Miller compensation)	3.0 pF	$g_{m1}/(2\pi\,\mathrm{UGB})$
R_c (zero nulling)	1.11 kΩ	$1/g_{m5,\mathrm{sim}} = 1/901.86\,\mu$
R_{sense} (CM averaging)	500 kΩ	$R_{sense} \gg R_{out2}$
$C_{L,CMFB}$ (Vcmfb node)	5.0 pF	$g_{m10}/(2\pi \times 8\,\mathrm{MHz})$
R_{OC} (DCOC integrator)	8.0 MΩ	τ/C_{OC}, $f_{HP} = 2$ kHz
C_{OC} (DCOC integrator)	10 pF	On-chip feasible
V_b (bias)	627 mV	$V_{th,N} + V_{ov0} = 477 + 150$
V_{ref} (CMFB reference)	0.9 V	$V_{DD}/2$

5.2 PGA Feedback Network

Table 4: PGA Resistor Ladder — Gain Settings (R1 = 10 kΩ)

Code	**Sw1**	**Sw2**	**Sw3**	**Gain (dB)**	**Gain (V/V)**	$R2_{eff}$
000	0	0	0	50 dB	316	3.16 MΩ (full ladder)
001	0	0	1	36.7 dB	68.3	683 kΩ
010	0	1	1	23.3 dB	14.6	146 kΩ
011	1	1	1	10 dB	3.16	31.6 kΩ

Table 5: Resistor Segment Values

Segment	Value	Shorted by
R_{seg0} (always in path)	31.6 kΩ	—
R_{seg1}	114.4 kΩ	Sw3 (lowest gain step)
R_{seg2}	537 kΩ	Sw2
R_{seg3}	2.477 MΩ	Sw1 (highest gain step)

Switch sizing: $W = 4\,\mu$m, $L = 180$ nm, expected $R_{on} \approx 1.2$ kΩ, gain error $\leq 3.7\%$ at 10 dB setting.

6 Simulation Results

6.1 4.1 DC Operating Point

For gain=50dB case,

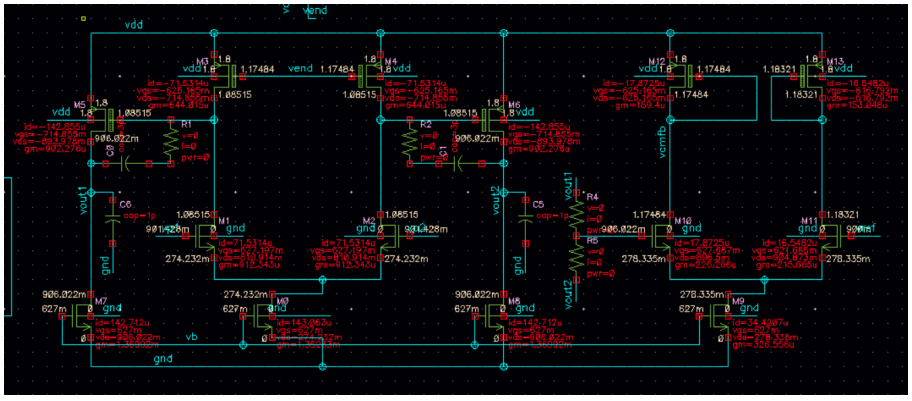


Figure 1: OTA DC Operating points

6.2 4.2 AC / Small-Signal Analysis

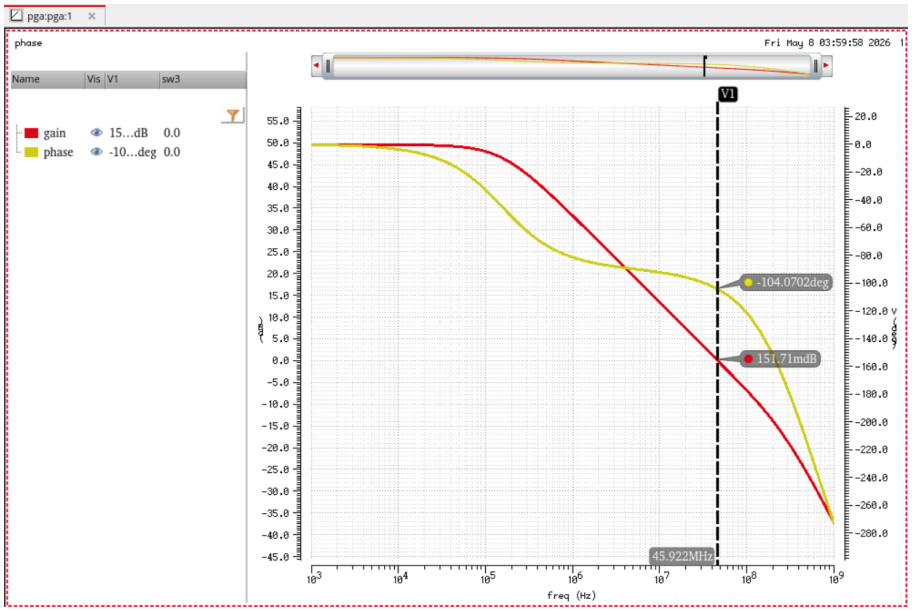


Figure 2: Differential gain, ugb, phase margin

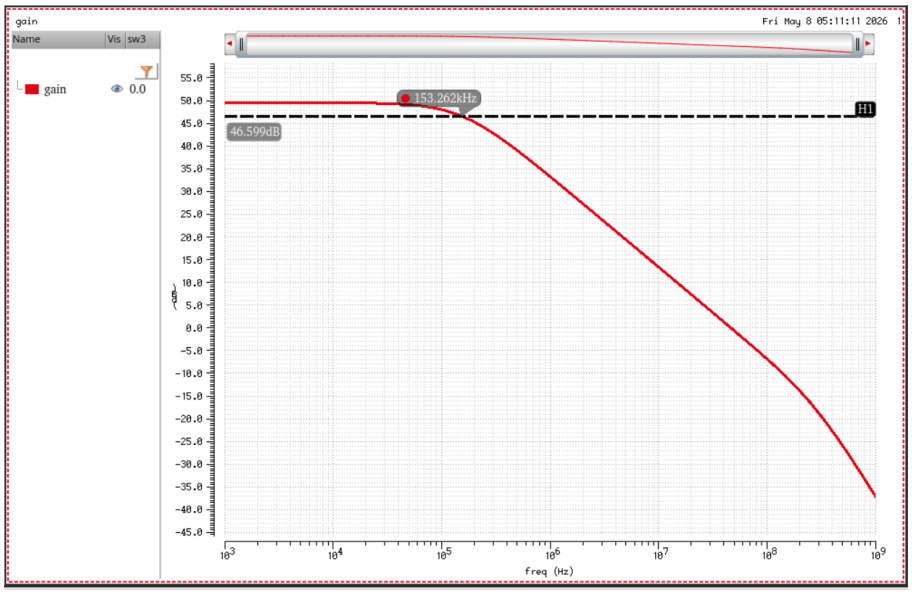


Figure 3: Differential gain f-3db

f3db	100.1K	> 75000
ugb	42.5M	
pm	85.6	> 60
dcgain	75.39	> 75
power	833.1u	< 1m

Figure 5: Enter Caption

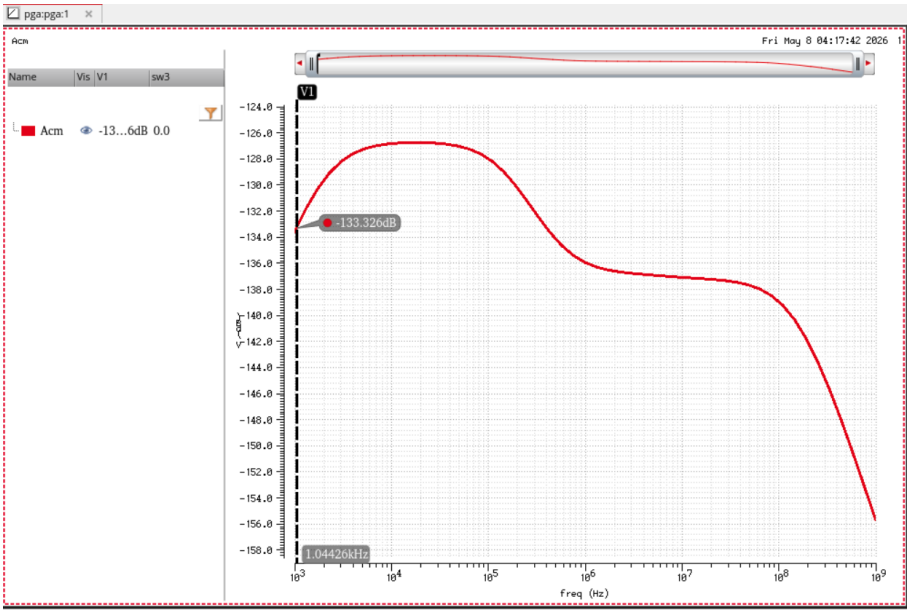


Figure 4: Acm(inputs in phase)

CMRR table:

Frequency	A_d	A_{cm}	**CMRR**
10 kHz	49.5 dB	−126.83 dB	176.33 dB
1 MHz	33.296 dB	−135.943 dB	169.239 dB
10 MHz	13.47 dB	−137.078 dB	150.548 dB

The decrease in CMRR at high frequencies is due to the reduction in tail node impedance caused by parasitic capacitances.

OTA:

6.3 4.3 Stability Analysis

Ckt designed for 50dB gain,

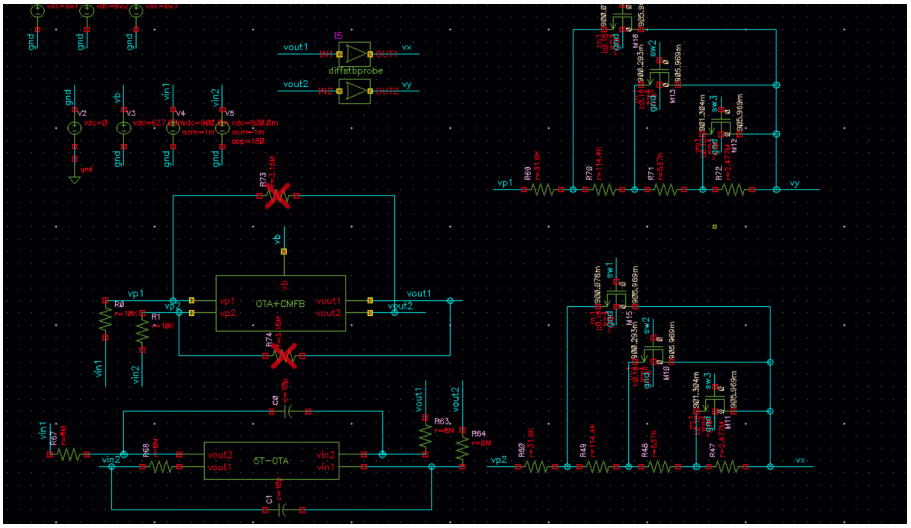


Figure 6: STB setup

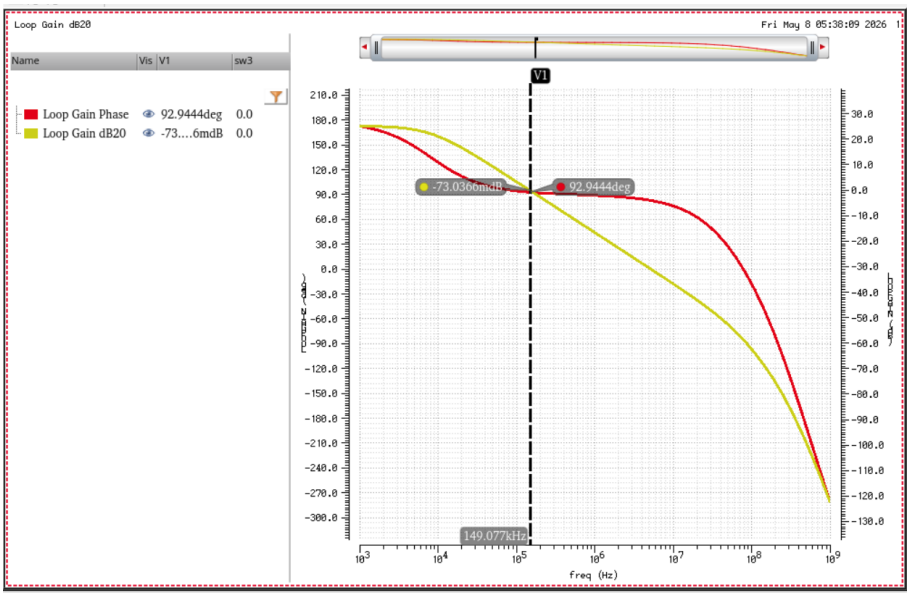


Figure 7: STB phase margin

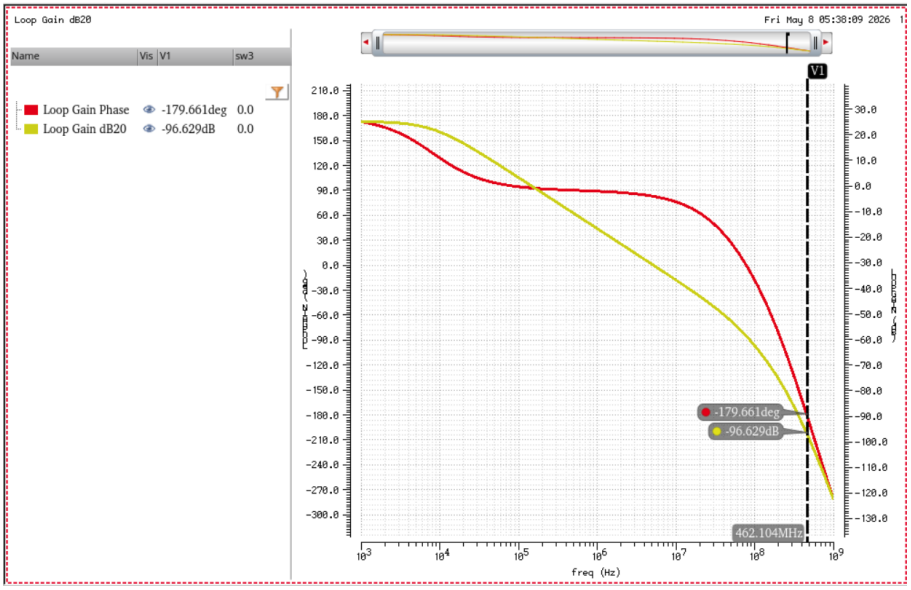


Figure 8: STB gain margin

Parameter	Target	Simulated
Differential loop gain	50 dB	49.53dB
Phase margin	60°–80°	92.9
Gain margin	> 10 dB	96.93dB

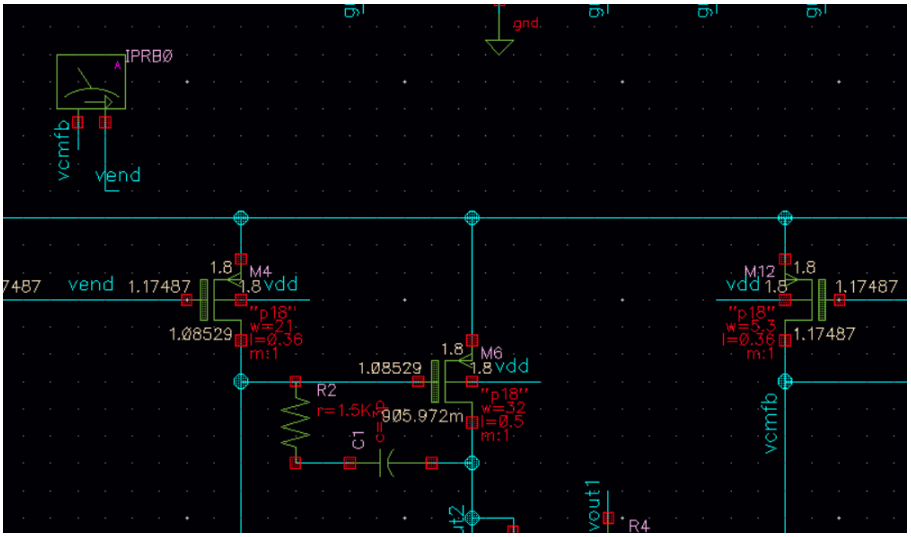


Figure 9: CMFB setup

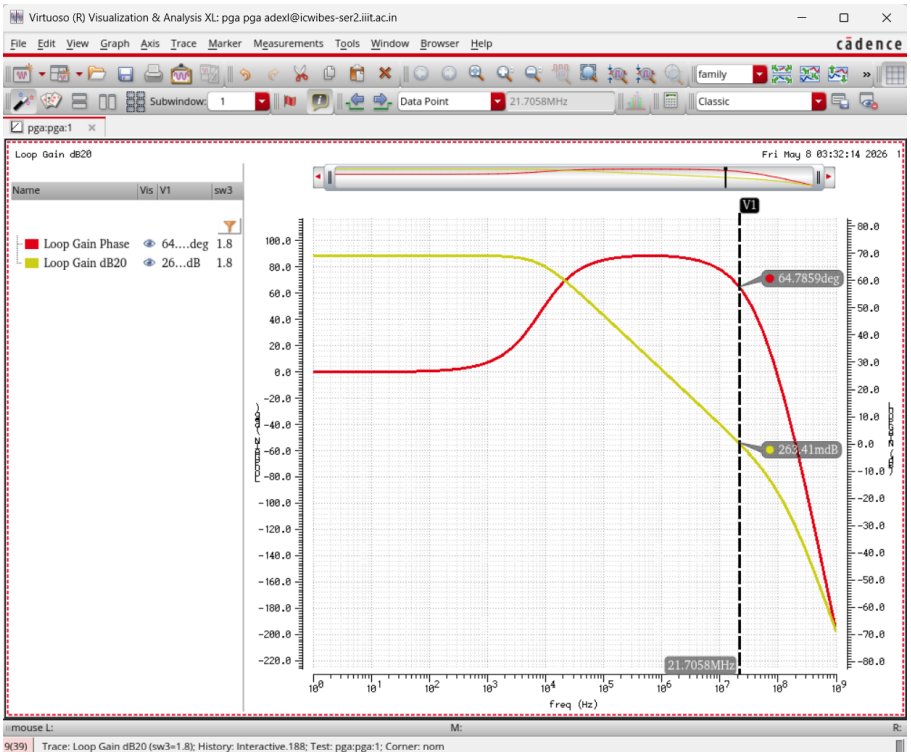


Figure 10: CMFB phase margin, ugb

Parameter	Target	Achieved
Loop gain	$> 40\,\text{dB}$	$\sim 69\,\text{dB}$
Phase margin (PM)	$> 60°$	$\sim 64°$
Unity-gain bandwidth (UGB)	5–10 MHz	$\sim 20\,\text{MHz}$

6.4 4.4 Transient / Large-Signal Analysis

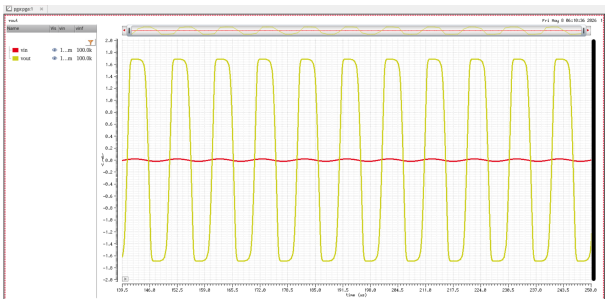


Figure 11: vin=10m, f=100k, clipped

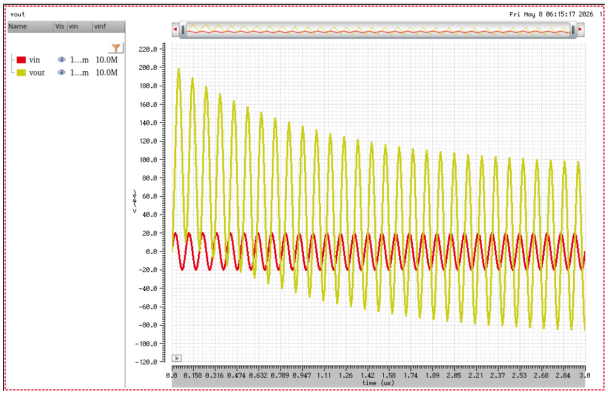


Figure 12: vin=10m, f=10M, poor gain

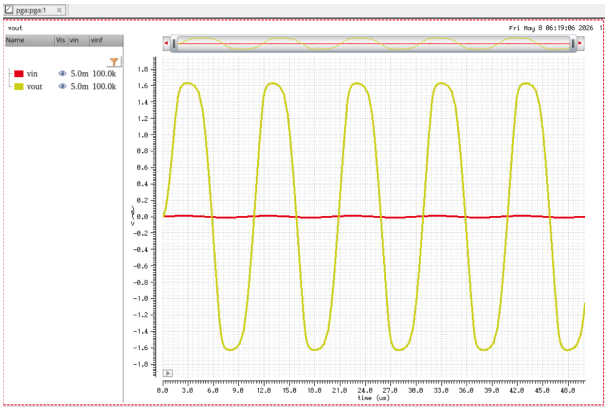


Figure 13: vin=5m, f=100k, no clipping, decent gain

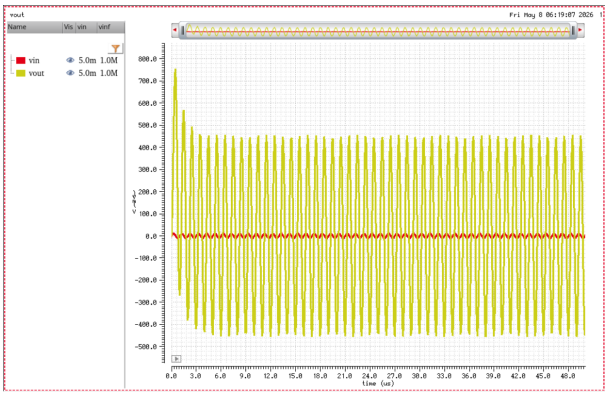


Figure 14: vin=5m f=1M, bad gain

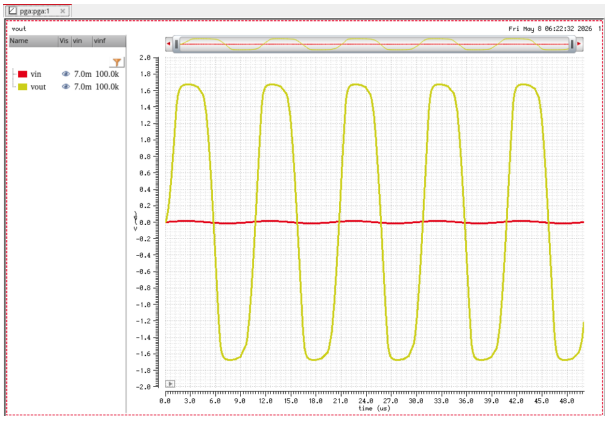


Figure 15: vin=7m f=100k, decent

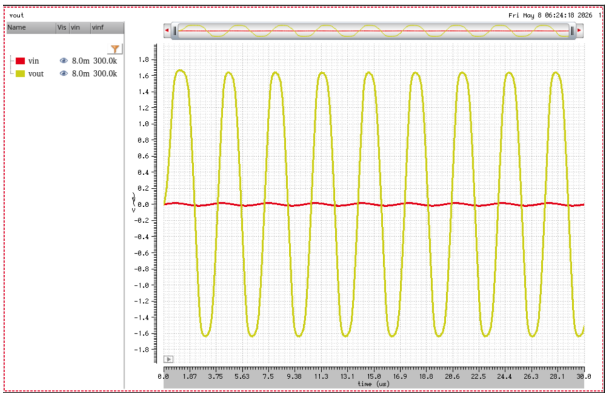


Figure 16: vin=8m f=400k, limits

Highest input that can be handled without clipping is vindiff=16m and highest frequency is 400k.

Input step= 10m Output step= 64m

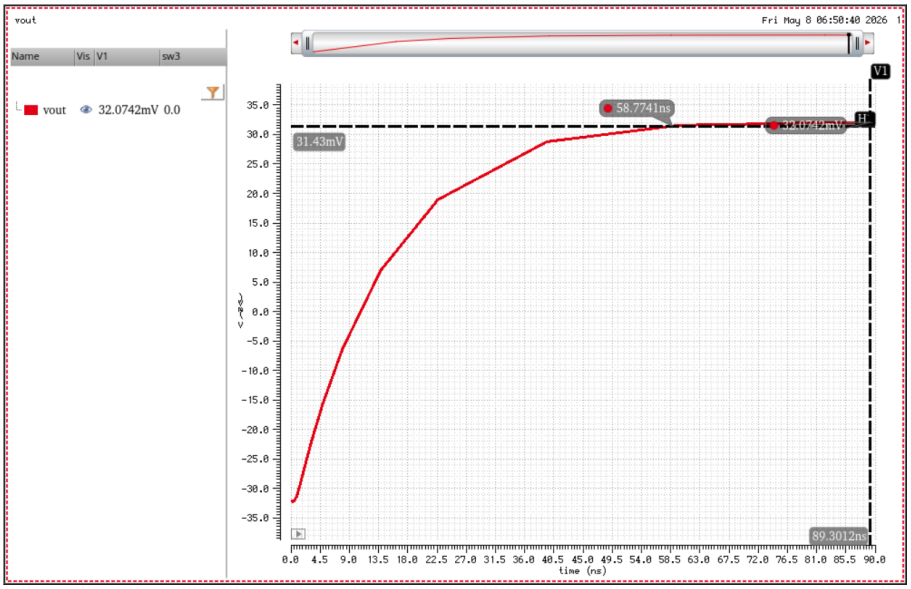


Figure 17: Settling time 2%

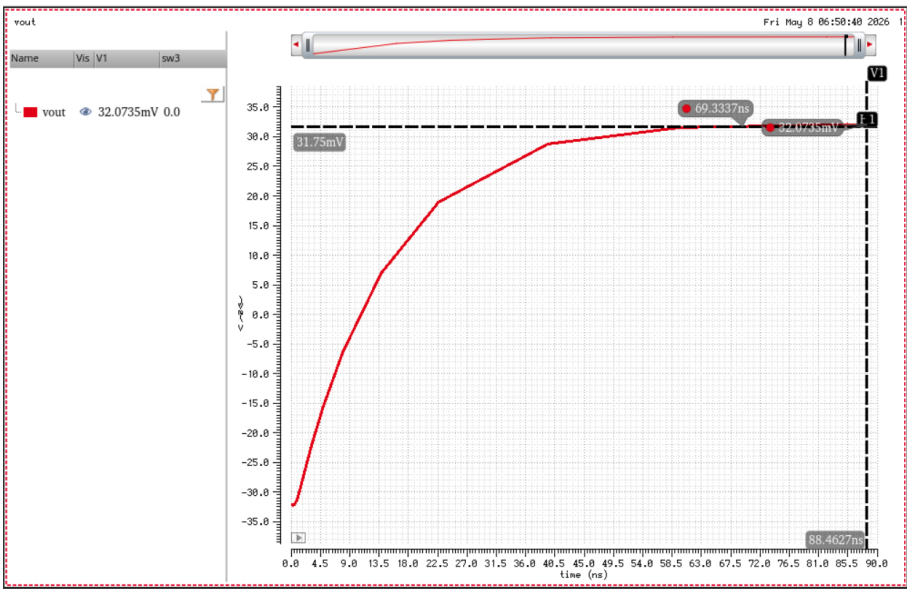


Figure 18: Settling time 1%

From the transient simulation, the output enters and remains within the 2% band ($32\,\text{mV} \pm 0.64\,\text{mV}$) at $t_s = 60\,\text{ns}$, and within the 1% band at $t_s = 69.33\,\text{ns}$, both well within the 200 ns target:

$$t_{s,2\%} = 58.77\,\text{ns} \tag{55}$$

$$t_{s,1\%} = 69.34\,\text{ns} \tag{56}$$

No overshoot between ts1 and ts2, phase margin is good.

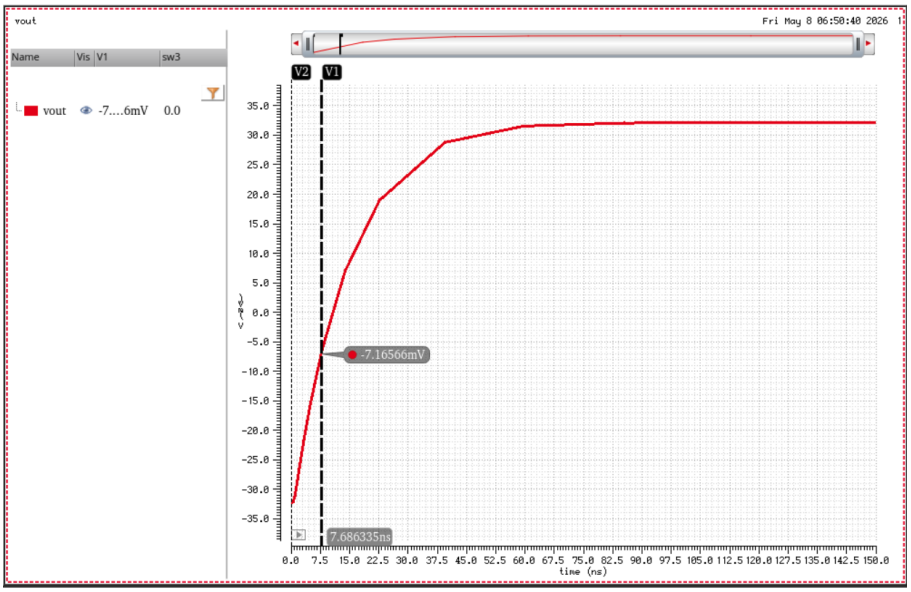


Figure 19: Slew rate closed loop

From the closed-loop transient waveform at 10 dB gain, the slew-limited region spans $\Delta V = 24.834\,\text{mV}$ over $\Delta t = 7.686\,\text{ns}$, giving a closed-loop slew rate of:

$$SR_{CL} = \frac{24.834\,\text{mV}}{7.686\,\text{ns}} = 3.23\,\text{V}/\mu\text{s} \tag{57}$$

The reduction from SR_{OL} is expected as negative feedback limits the rate of change seen at the output in closed loop.

6.5 4.5 Noise Analysis

[H]

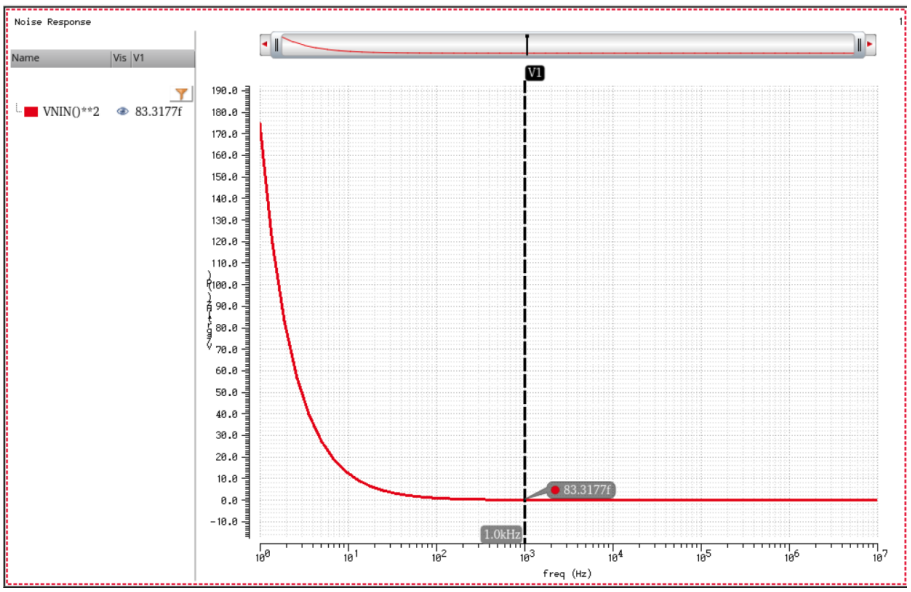


Figure 21: vnin at 1k

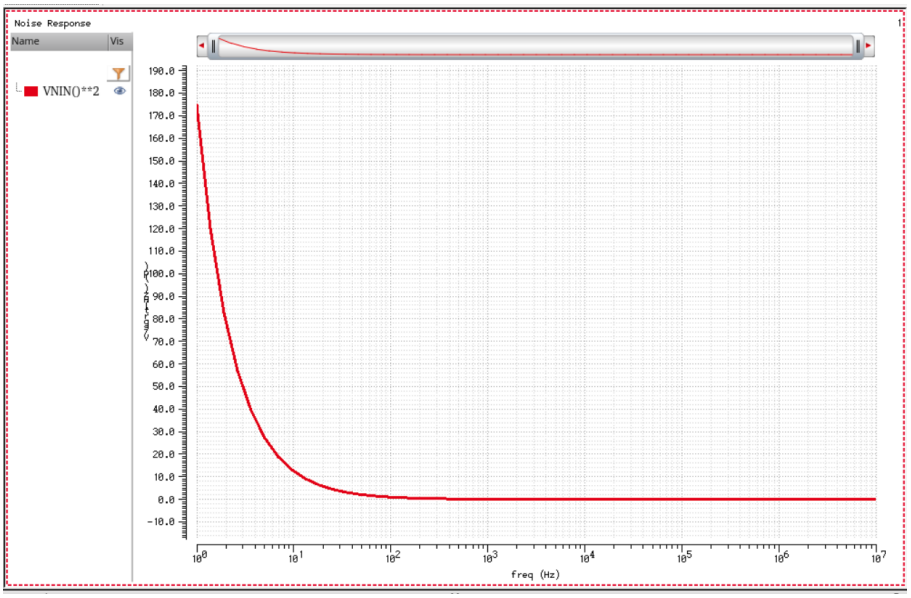


Figure 20: vnin

integrated=329.7m

6.6 4.6 DC Offset Cancellation

Offset: 5.968m The DCOC loop is enabled after measuring the open-loop offset. The high-pass filter:

$$f_{HP} = \frac{1}{2\pi R_{OC} C_{OC}} = \frac{1}{2\pi \times 8\,\mathrm{M} \times 10\,\mathrm{p}} = 1.99\ \mathrm{kHz} \approx 2\ \mathrm{kHz} \tag{58}$$

Convergence time $\approx 5\tau = 5 \times 79.6\,\mu\text{s} = 398\,\mu\text{s}$. Target: output DC offset $\leq 10\,\text{mV}$ after cancellation.

6.7 4.7 Programmable Gain Verification

Table 6: PGA Gain Verification — Measured vs. Target

Code	sw1	sw2	sw3	Target (dB)	Meas. (dB)	Gain (V/V)	$f_{-3\text{dB}}$	Error
1	0	0	0	50	49.53	298.4	158.5 kHz	0.94%
2	0	0	1	36.7	36.58	67.5	712 kHz	0.33%
3	0	1	0	23.3	23.29	14.6	3.13 MHz	0.04%
4	1	0	0	10	10.13	3.21	11.52 MHz	1.30%

The inverse gain–bandwidth relationship confirms the OTA unity-gain bandwidth remains constant at $\approx 48\,\text{MHz}$ across all settings:

$$\text{UGB} \approx A_{CL} \times f_{-3\text{dB}}: \quad 298.4 \times 158.5\,\text{kHz} = 47.3\,\text{MHz}, \quad 3.21 \times 11.52\,\text{MHz} = 37.0\,\text{MHz} \tag{59}$$

7 Summary: Target vs. Achieved

Table 7: Complete Specification Comparison — Target vs. Simulated

Parameter	**Target**	**Simulated**
Core OTA		
DC open-loop gain	$\geq 70\,\text{dB}$	$75.39\,\text{dB}$
$-3\,\text{dB}$ BW (CL @ $50\,\text{dB}$)	$\geq 750\,\text{kHz}$	$100\,\text{kHz}$
UGB	—	$42.5\,\text{MHz}$
Phase margin	$60°$–$85.6°$	$71.4°$
Slew rate	$\geq 20\,\text{V}/\mu\text{s}$	$40\,\text{V}/\mu\text{s}$
Settling time (2%)	$\leq 200\,\text{ns}$	$\approx 60\,\text{ns}$
Power (OTA+CMFB)	$\leq 1\,\text{mW}$	$853\text{u}\,\mu\text{W}$
CMFB		
CMFB loop gain	$\geq 40\,\text{dB}$	$69\,\text{dB}$
CMFB UGF	5–$10\,\text{MHz}$	$20\,\text{MHz}$
CMFB PM	$\geq 60°$	64
PGA		
Gain range	10–$50\,\text{dB}$	10–$50\,\text{dB}$
Gain error	$\leq 5\%$	$\leq 1.3\%$
Total power (incl. DCOC)	$< 1.5\,\text{mW}$	$\approx 1.2m\,\mu\text{W}$

References

[1] P. E. Allen and D. R. Holberg, *CMOS Analog Circuit Design*, Oxford University Press.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, McGraw-Hill.

[3] S. D'Amico *et al.*, "A low-power analog baseband section for 60-GHz receivers in 90-nm CMOS," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 8, pp. 1724–1735, Aug. 2014.

[4] W. Wu, L. Zhang, L. Zhang, and Y. Wang, "A temperature-robust analog baseband chain for automotive radar," in *Proc. EDSSC*, Hsinchu, Taiwan, 2017, pp. 1–2.